

Final demo :)

This is a demo version of the final exam. Final will consist of 6 problems: generating functions in probability, Polya-Redfield counting, planarity and Euler formula, graph colouring, linearity of expected value and one old topic. Some problems will be similar to demo 

You may use one A4 cheat sheet in any form handwritten, printed, aquarelle painting...

1. [old topic] Write the Taylor expansion at $x_0 = 0$ of the function

$$f(x) = (1 - 5x^2)^{1/2}.$$

2. [generating functions in probability] I throw a coin until the sequence HTT appears. The coin is not fair and shows H with probability 0.4.

- Draw the directed graph for this game.
- Setup a system of equations for the sequences with given endpoint and starting point at the start of the game.
- Explain how would you calculate expected number of coin tosses from the start till the end of the game.

Note: you don't need to do actual calculations nor to solve the system, just describe how can you extract the expected value using derivatives.

3. [Polya-Redfield counting] Consider a 2×2 mosaic of gems, where each gem may be of one of k possible types. We consider mosaics identical if the can be transformed one into another by rotation or flip.

- List all elements of the group of actions that act on mosaics.
- Write each action as a combination of cycles and provide its cyclic monomial.
- Find the cyclic index and the number of different possible colourings.

4. [planarity and Euler formula] The convex polyhedron consists of only of triangular and hexagonal faces faces with with three faces meeting at each vertex.

How many triangular faces are there?

5. [graph colouring] Consider the graph with two components. The first component is a K_4 graph, the second one — C_4 graph. Components are non connected between them. You have k colours available.

In how many ways can you color the vertices of this graph if adjacent colors can't be coloured in the same colour?

6. [linearity of expected value] There are 20 types of desserts in a café. In the queue there are 10 clients. Each client prefers one of the desserts at random with equal probabilities.

- What is the average number of dessert types bought?
 - What is the average number of dessert types bought if each client will not buy any dessert with probability 0.3?
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